Reciprocal Summability of Every Unbounded Natural Collatz Orbit: A Lean Proof

Collatz proof project

September 5, 2026

Abstract

Every unbounded positive natural orbit of the shortcut Collatz map has a convergent series of reciprocal values. We prove this in Lean by combining an equal-time injectivity argument with existing binomial and growth bounds. An explicit finite packing inequality supplies a geometric shell majorant. Consequently every such orbit has bounded ideal correction, and its inverse multiplicative drift tends to zero. This closes the correction-coverage gap previously recorded in this repository; it does not exclude unbounded orbits, prove a strict density gap, or rule out nontrivial cycles. We make no claim of mathematical novelty.

1 Statements and conventions

Let $T(n) = n/2$ for even n and $(3n+1)/2$ for odd n. Set $x_t = T^t(N)$, and let j_t count odd values among $x_0, \dots, x_{t-1}$. Define the nonnegative ideal correction $c_t(N)$ by the exact identity

$$x_t = \frac{3^{j_t}}{2^t}\big(N + c_t(N)\big).$$

In the repository, Supercritical(N) means only that $c_t(N)$ is bounded above as t varies. It does not mean a strict density gap above $\log 2/\log 3$.

Theorem 1. *If $N > 0$ and $\forall B\, \exists t\ (B < x_t)$, then*

$$\sum_{t\geq 0} \frac{1}{x_t} < \infty, \qquad \sup_t c_t(N) < \infty, \qquad \frac{2^t}{3^{j_t}} \longrightarrow 0.$$

For positive seeds, bounded ideal correction is equivalent to unboundedness of the natural orbit.

An unbounded deterministic orbit cannot repeat a state. Moreover, T^k is injective on its entire orbit set for every fixed k: coalescence at equal times would imply a repetition at two later times.

2 Finite packing with explicit constants

Theorem 2. *Let $S \subseteq [0, 32^m) \cap \mathbb{N}$ be finite, and suppose T^{5m} is injective on S. Then*

$$8^m|S| \leq 2 \cdot 216^m + 243^m.$$

Put $k = 5m$. For any $n < 2^k$, the existing upper master inequality gives

$$2^k\big(T^k(n) + 1\big) \leq 3^{j_k(n)}\big(n + 2^{k - j_k(n)}\big), \qquad T^k(n) < 2 \cdot 3^{j_k(n)}.$$

Split S at odd count $3m$. Low-count elements map into $[0, 2 \cdot 27^m)$; injectivity bounds their number by $2 \cdot 27^m$. The high-count elements are bounded by all residues with at least $3m$ odd steps. The verified parity bijection gives exactly $\binom{k}{j}$ residues with j odd steps. Therefore the weighted binomial identity gives

$$2^{3m} \sum_{j \geq 3m} \binom{5m}{j} \leq \sum_j 2^j \binom{5m}{j} = 3^{5m}.$$

Multiplying the low-count bound by 8^m and adding proves the inequality. This applies to every finite subset of an unbounded orbit below 32^m, including values reached at arbitrarily late times.

3 Geometric shells and time summability

For orbit values in $[32^m, 32^{m+1})$, the packing bound at scale $m+1$ and $1/n \le 32^{-m}$ give

$$\sum_{n \text{ in the shell}} \frac{1}{n} \le b_m := 54\left(\frac{27}{32}\right)^m + \frac{243}{8}\left(\frac{243}{256}\right)^m.$$

Both ratios are less than one. Inductively splitting any finite set of positive orbit values at the shell endpoints bounds its reciprocal sum by an initial partial sum of $\sum_m b_m$.

For a finite time prefix, take the image of its orbit values as a finite set. Unboundedness supplies injectivity in time, so passing to this set preserves the reciprocal sum without losing multiplicities. Each such sum is bounded by $\sum_m b_m$, proving summability. The displayed majorant has sum less than 1000; the named Lean theorem asserts summability rather than this numerical constant.

This distinction between values and time is essential. A finite cycle has a finite reciprocal sum over its distinct values but a divergent reciprocal series over its infinite time sequence. The packing step alone would not justify summability for a repeating orbit.

4 Correction coverage and drift

The earlier module `Reciprocal.lean` proves, for positive seeds,

$$\text{Supercritical}(N) \iff \text{the time-indexed reciprocal series is summable.}$$

It also proves that bounded correction forces escape to infinity. Applying these results gives both directions of the new equivalence between bounded correction and unboundedness.

Finally, $c_t(N)$ tends to its finite supremum and $1/x_t$ tends to zero. The identity

$$\frac{2^t}{3^{j_t}} = \big(N + c_t(N)\big)\frac{1}{x_t}$$

proves the last limit in Theorem 1. Equivalently, on the additive logarithmic scale, $j_t \log 3 - t \log 2$ tends to positive infinity. The Lean declaration uses the inverse-drift limit directly. Dividing by t need not leave a positive limit: no uniform strict density gap is claimed.

5 Formal verification and provenance

The new modules are `Collatz/OrbitPacking.lean` and `Collatz/OrbitSummability.lean`. Principal declarations include `unbounded_orbit_reciprocal_summable`, `unbounded_orbit_supercritical`, `supercritical_iff_unbounded_orbit`, and `unbounded_inverse_drift_tendsto_zero`. The build and selected audit pass: 113 declarations use only standard foundations. Three exact finite controls check full small residue images, the geometric identity, and the coalescence/multiplicity boundary. The audit is dependency inspection, not independent kernel replay.

A literature check prompted this route. Garcia and Tal, *A note on the generalized $3n+1$ problem*, Acta Arithmetica 90(3) (1999), 245–250, doi:10.4064/aa-90-3-245-250, give a power-saving interval estimate in equation (6), with orbit inclusion in Corollary 1. That quantitative estimate implies reciprocal summability; Banach density zero alone would not suffice. A July 2026 MathOverflow question points out this implication. We checked the original six-page paper. The present Lean proof uses a simpler origin-based packing argument with explicit constants from this repository, importing neither Garcia–Tal's theorem nor its cited Heppner input.

6 What remains

The earlier ledger's possible unbounded orbits with divergent reciprocal series are now excluded. All hypothetical unbounded positive orbits lie within the bounded-correction framework. This gives no contradiction within that framework, no seed-independent absolute bound on all shifted ideal corrections, and no exclusion of nontrivial cycles. The full Collatz conjecture remains unresolved.